

A Step-by-Step Repair of the Killed-Resolvent Proof

Why the old boundary argument fails and why the corrected one works

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Abstract

The current paper reaches the right-looking 3×3 linear system by using one function in two incompatible ways. It first treats the resolvent as zero below the liquidation barrier, but then applies a full-line exponential integral to its particular and homogeneous pieces. This note explains the problem using the constant particular solution $1/q$, introduces the required notation from scratch, and gives a corrected proof. The repair is simple in principle: a downward jump integral must stop at the barrier. For an exponential term, this truncated integral equals the familiar rational term minus one extra exponential term. The last two rows of the 3×3 system make these extra terms cancel; the first row imposes absorption at the barrier. The numerical matrix can therefore be retained, but the old boundary interpretation must be replaced.

1 What the proof is trying to calculate

Fix a moment order k and a time-Laplace variable q . The object of interest is the Laplace transform of a payoff that is set to zero after the log health factor crosses its liquidation boundary. In the paper's original coordinate, the log-ratio process is denoted by z and liquidation occurs at

$$a := -h_0.$$

The resolvent is needed only for $z > a$, but the generator contains jump integrals whose landing points can lie below a . This is where the boundary treatment matters.

It is clearer to measure distance from liquidation. Define

$$x := z - a = z + h_0. \tag{1}$$

Then $x > 0$ means that the position is alive, $x = 0$ is the barrier, and $x < 0$ means that a downward jump has crossed the barrier. Let

$$u(q, x)$$

denote the killed resolvent in this translated coordinate. Absorption means

$$u(q, x) = 0, \quad x \leq 0. \tag{2}$$

2 Why the current proof does not work

Consider one downward exponential jump phase with rate $r > 0$. If jump size is $y > 0$, then a jump from x lands at $x - y$. Applied to the killed function, the downward jump operator is

$$K_{r,-}^D u(q, x) := \int_0^\infty u(q, x - y) r e^{-ry} dy = \int_0^x u(q, x - y) r e^{-ry} dy. \quad (3)$$

The second equality follows from (2): terms with $y \geq x$ land at or below zero and contribute nothing. Hence

$$\lim_{x \downarrow 0} K_{r,-}^D u(q, x) = 0. \quad (4)$$

The old proof then writes the interior formula as

$$u(q, x) = P(q, x) + H(q, x), \quad x > 0,$$

where P is a particular solution and H is a sum of characteristic exponentials. It uses linearity to replace (4) by

$$K_{r,-}[P](0) + K_{r,-}[H](0) = 0,$$

and evaluates these two terms using full-line exponential formulas. That step is not valid: P and H have only been introduced as pieces of an *interior* formula. Their exponential formulas extend them to $x < 0$, whereas the killed function in (2) is zero.

The contradiction is visible for the constant $1/q$

For the survival probability, the particular solution inside the alive region is the constant

$$P(q, x) = \frac{1}{q}, \quad x > 0.$$

There are now two different calculations:

$$\begin{aligned} \text{killed extension: } K_{r,-}^D[P](0^+) &= 0, \\ \text{constant extended to the whole line: } \int_0^\infty \frac{1}{q} r e^{-ry} dy &= \frac{1}{q}. \end{aligned}$$

Both calculations are individually correct, but they concern different functions below the barrier. The old proof uses the first interpretation to state the boundary condition and the second interpretation to compute its components. Linearity cannot connect two different extensions.

There is a second problem. For a characteristic term $e^{\gamma x}$, the full-line calculation requires

$$\int_0^\infty e^{-\gamma y} r e^{-ry} dy < \infty, \quad \text{or equivalently} \quad \operatorname{Re}(\gamma) > -r.$$

Some negative characteristic roots can lie to the left of the pole $-r$. Then the integral diverges even though the rational expression $r/(r + \gamma)$ is finite. A proof based on the literal full-line integral therefore cannot cover all roots used by the matrix.

The failure is thus not that the final matrix necessarily has the wrong numbers. The failure is that the stated boundary condition does not imply those matrix rows by the reasoning given.

3 Notation for the corrected proof

The main symbols are collected here before the calculation begins.

Symbol	Meaning
z	Original log-ratio state used in the paper.
$a = -h_0$	Liquidation boundary in the z coordinate.
$x = z + h_0$	Distance from the liquidation boundary; the alive region is $x > 0$.
$u(q, x)$	Killed resolvent, set equal to zero for $x \leq 0$.
$v(q, x)$	Exponential formula proposed only on the interior $x > 0$.
$r_{d,-}^{(k)}$	Rate of downward exponential phase $d \in \{1, 2\}$ under the k -tilted measure.
$\lambda_{d,-}^{(k)}$	Intensity of downward phase d ; it is positive.
$K_{r,-}^D$	True killed downward operator; its integral stops at jump size x .
$\mathcal{L}^{(k),D}$	True generator of the process killed at $x = 0$.
$\mathcal{L}^{(k),\text{rat}}$	Algebraic “rational” operator used to identify particular solutions and characteristic roots.
$C_r[v]$	Coefficient of the extra e^{-rx} term created by truncating a downward integral.

4 The true killed generator

Fix a tilt order k . Under the k -tilted measure, the log-ratio process has two upward exponential jump phases and two downward exponential jump phases. Index upward phases by $a \in \{1, 2\}$ and downward phases by $d \in \{1, 2\}$. Write their positive intensities as $\lambda_{a,+}^{(k)}$ and $\lambda_{d,-}^{(k)}$, and their positive rates as $r_{a,+}^{(k)}$ and $r_{d,-}^{(k)}$. The exact intensity formulas are not needed for the boundary argument; positivity and the two distinct downward rates are what matter.

More explicitly, phase 1 comes from jumps of asset 1 and phase 2 from jumps of asset 2. Since $Z = X_1 - X_2$, a negative jump of asset 2 is an upward jump of Z , while a positive jump of asset 2 is a downward jump of Z . The rates and intensities are

$$\begin{aligned}
 r_{1,+}^{(k)} &= \frac{1}{\eta_{1,+}}, & \lambda_{1,+}^{(k)} &= \lambda_1 p_1, \\
 r_{2,+}^{(k)} &= \frac{1}{\eta_{2,-}} + k, & \lambda_{2,+}^{(k)} &= \frac{\lambda_2(1-p_2)}{1+k\eta_{2,-}}, \\
 r_{1,-}^{(k)} &= \frac{1}{\eta_{1,-}}, & \lambda_{1,-}^{(k)} &= \lambda_1(1-p_1), \\
 r_{2,-}^{(k)} &= \frac{1}{\eta_{2,+}} - k, & \lambda_{2,-}^{(k)} &= \frac{\lambda_2 p_2}{1-k\eta_{2,+}}.
 \end{aligned}$$

The condition $k\eta_{2,+} < 1$ makes the last rate and intensity positive.

For a function defined on $x > 0$, the generator of the killed process is

$$\begin{aligned} \mathcal{L}^{(k),D} f(x) &= \mu^{(k)} f'(x) + \frac{\sigma_Z^2}{2} f''(x) + \sum_{a=1}^2 \lambda_{a,+}^{(k)} (K_{a,+}^{(k)} f(x) - f(x)) \\ &\quad + \sum_{d=1}^2 \lambda_{d,-}^{(k)} (K_{d,-}^{(k),D} f(x) - f(x)), \quad x > 0. \end{aligned} \quad (5)$$

Here all phase intensities are positive, and

$$K_{a,+}^{(k)} f(x) := \int_0^\infty f(x+y) r_{a,+}^{(k)} e^{-r_{a,+}^{(k)} y} dy, \quad (6)$$

$$K_{d,-}^{(k),D} f(x) := \int_0^x f(x-y) r_{d,-}^{(k)} e^{-r_{d,-}^{(k)} y} dy. \quad (7)$$

The upper limit x in (7) is essential: a downward jump larger than x crosses the liquidation barrier, where the payoff is zero.

The drift $\mu^{(k)}$ and diffusion variance σ_Z^2 are the corresponding tilted quantities. In particular, $\sigma_Z > 0$ means that absorption requires the ordinary boundary condition $u(q, 0) = 0$.

The paper's rational characteristic function $\psi_Z^{(k)}(\beta)$ is useful, but it must be interpreted carefully. It is obtained by replacing each phase operator acting on $e^{\beta x}$ by a rational multiplier. It is an algebraic device for finding exponentials that solve the interior equation. It is not the result of applying the killed operator at the boundary.

5 The missing truncation identity

Lemma 1 (Truncated downward phase). *Let $r > 0$ and let $\beta \in \mathbb{C} \setminus \{-r\}$. On $x > 0$,*

$$\int_0^x e^{\beta(x-y)} r e^{-ry} dy = \frac{r}{r+\beta} e^{\beta x} - \frac{r}{r+\beta} e^{-rx}. \quad (8)$$

The identity holds for every such β ; no condition $\operatorname{Re}(\beta) > -r$ is required because the integral is over the finite interval $[0, x]$.

Proof. Direct integration gives

$$e^{\beta x} r \int_0^x e^{-(r+\beta)y} dy = e^{\beta x} \frac{r}{r+\beta} (1 - e^{-(r+\beta)x}),$$

which is (8). □

For a finite exponential sum

$$v(x) = \sum_{\beta \in \mathcal{E}} a_\beta e^{\beta x},$$

define its *truncation-correction coefficient*

$$C_r[v] := \sum_{\beta \in \mathcal{E}} a_\beta \frac{r}{r+\beta}. \quad (9)$$

This is simply a number calculated from the coefficients of the interior formula v . It is not the value of a jump integral at the boundary. Lemma 1 gives

$$K_{r,-}^D v(x) = \sum_{\beta \in \mathcal{E}} a_\beta \frac{r}{r+\beta} e^{\beta x} - C_r[v] e^{-rx}. \quad (10)$$

The first term is the expression produced by the rational calculation used in the paper. The second term is what that calculation omits when jumps crossing the barrier are killed. This omitted term is the key to the repair.

For example, if $v(x) = 1/q$, then $C_r[v] = 1/q$ and

$$K_{r,-}^D v(x) = \frac{1}{q} - \frac{1}{q} e^{-rx}. \quad (11)$$

This tends to zero as $x \downarrow 0$, as it must. The old proof kept the first $1/q$ but did not explicitly derive the cancelling second term.

6 The proposed repair

The proposal has four steps.

1. Work only on the alive half-line $x > 0$.
2. Use the rational equation to construct an interior candidate $v(q, x)$.
3. Apply the *true truncated* downward operators to this candidate. This produces two extra terms, one proportional to $e^{-r_{1,-}^{(k)} x}$ and one proportional to $e^{-r_{2,-}^{(k)} x}$.
4. Choose the three homogeneous coefficients so that the value at the barrier is zero and both extra terms vanish.

The next proposition carries out these steps.

Let $k \geq 1$ be the integer moment order, let $b \in (0, 1)$ be the liquidation threshold parameter, and let q be the time-Laplace variable. Translation removes h_0 from the forcing:

$$G_k(x) := (e^x - b)^k = \sum_{j=0}^k d_{k,j} e^{jx}, \quad d_{k,j} := \binom{k}{j} (-b)^{k-j}.$$

Assume that all exponentials in the forcing are in the positive-moment domain. In the bivariate Kou model this requires, in particular,

$$k\eta_{1,+} < 1, \quad k\eta_{2,+} < 1. \quad (12)$$

The second condition makes the k -tilt well defined; the first ensures that the long-leg exponential moments and $\psi_Z^{(k)}(j)$ for $j \leq k$ are finite.

Let q initially lie to the right of the time-Laplace abscissa of convergence and suppose $q \neq \psi_Z^{(k)}(j)$ for $j = 0, \dots, k$. Define the algebraic particular solution

$$P_k(q, x) := \sum_{j=0}^k \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} e^{jx}. \quad (13)$$

Suppose the rational characteristic equation

$$\psi_Z^{(k)}(\gamma) = q$$

has three simple roots $\gamma_4, \gamma_5, \gamma_6$ with negative real part, none equal to a phase pole. Consider

$$v_k(q, x) := P_k(q, x) + \sum_{m=4}^6 B_m^{(k)}(q) e^{\gamma_m(q)x}, \quad x > 0. \quad (14)$$

Proposition 1 (Correct barrier conditions). *Assume $\sigma_Z > 0$ and $r_{1,-}^{(k)} \neq r_{2,-}^{(k)}$. The function*

$$\widehat{U}_k(q, x) := \begin{cases} v_k(q, x), & x > 0, \\ 0, & x \leq 0, \end{cases} \quad (15)$$

solves

$$(q - \mathcal{L}^{(k),D}) \widehat{U}_k(q, x) = G_k(x), \quad x > 0, \quad (16)$$

with an absorbing boundary if and only if

$$v_k(q, 0) = 0, \quad C_{r_{1,-}^{(k)}}[v_k] = 0, \quad C_{r_{2,-}^{(k)}}[v_k] = 0. \quad (17)$$

These conditions give the 3×3 system

$$\begin{pmatrix} 1 & 1 & 1 \\ \frac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_4} & \frac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_5} & \frac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_6} \\ \frac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_4} & \frac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_5} & \frac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_6} \end{pmatrix} \begin{pmatrix} B_4^{(k)} \\ B_5^{(k)} \\ B_6^{(k)} \end{pmatrix} = - \begin{pmatrix} P_k(q, 0) \\ C_{r_{1,-}^{(k)}}[P_k] \\ C_{r_{2,-}^{(k)}}[P_k] \end{pmatrix}, \quad (18)$$

where

$$C_r[P_k] = \sum_{j=0}^k \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \frac{r}{r + j}. \quad (19)$$

Thus (18) is algebraically the same system as in the paper, after substituting $x = z + h_0$.

Proof. Apply the killed generator (5) to the interior ansatz (14). Upward phase integrals are not truncated by the lower barrier. For the positive forcing exponents they converge under (12); for the negative characteristic roots they converge automatically. Their action is therefore the usual rational-symbol action.

For each downward phase, use (10). Define $\mathcal{L}^{(k),\text{rat}}$ to be the algebraic operator that keeps the first term in (10) but not its truncation correction. It is exactly the operator represented by the rational function $\psi_Z^{(k)}$. For $x > 0$,

$$\mathcal{L}^{(k),D} v_k(x) = \mathcal{L}^{(k),\text{rat}} v_k(x) - \sum_{d=1}^2 \lambda_{d,-}^{(k)} C_{r_{d,-}^{(k)}}[v_k] e^{-r_{d,-}^{(k)} x}. \quad (20)$$

By construction of P_k and the characteristic roots,

$$(q - \mathcal{L}^{(k),\text{rat}}) v_k(x) = G_k(x).$$

It follows from (20) that

$$(q - \mathcal{L}^{(k),D})v_k(x) = G_k(x) + \sum_{d=1}^2 \lambda_{d,-}^{(k)} C_{r_{d,-}^{(k)}}[v_k] e^{-r_{d,-}^{(k)}x}. \quad (21)$$

Because the two rates are distinct and their intensities are positive, the two extra exponentials are linearly independent. They cannot cancel each other for every $x > 0$. Equation (16) therefore holds exactly if and only if each correction coefficient is zero:

$$C_{r_{1,-}^{(k)}}[v_k] = C_{r_{2,-}^{(k)}}[v_k] = 0.$$

Since $\sigma_Z > 0$, the lower boundary is regular for the Brownian component and killing imposes the Dirichlet condition $v_k(q, 0) = 0$. These are the three conditions in (17). Applying them to (14) gives (18) and (19). Standard uniqueness of the killed resolvent in the relevant exponential-growth class then identifies the candidate with the probabilistic resolvent. $\square$

7 Survival probability

For survival, set $k = 0$ and $G_0 \equiv 1$. The algebraic particular solution is $P_0 = 1/q$, hence

$$P_0(0) = \frac{1}{q}, \quad C_r[P_0] = \frac{1}{q}.$$

The right-hand side of the survival system is therefore $-(1/q, 1/q, 1/q)^\top$, exactly as in the paper. The correct explanation is not that the downward convolution of the killed resolvent at the barrier is $1/q$; it is that the constant *interior particular solution* produces a barrier-truncation residual of size $(1/q)e^{-rx}$, which the homogeneous terms must cancel.

8 Why the proposal works while the old argument does not

There are five concrete differences.

1. *Below the barrier.* The current proof uses the zero extension when stating the boundary condition, but full-line exponential extensions when evaluating its pieces. The corrected proof never extends the interior candidate below the barrier.
2. *The constant particular solution.* The current proof combines a zero killed-boundary integral with the full-line identity $K[1/q] = 1/q$. The corrected proof instead derives

$$K^D[1/q](x) = q^{-1}(1 - e^{-rx}),$$

which tends to zero at the barrier without contradiction.

3. *Roots beyond a phase pole.* The full-line integral can diverge when $\text{Re}(\gamma) \leq -r$. The corrected integral is over the finite interval $[0, x]$ and is valid for every $\gamma \neq -r$.
4. *Meaning of the matrix.* The current proof presents the last two rows as jump-operator values at the boundary. The corrected proof shows that they cancel the coefficients of the two omitted functions $e^{-r_{1,-}^{(k)}x}$ and $e^{-r_{2,-}^{(k)}x}$.

5. *Direct verification.* The corrected proof substitutes the candidate into the true generator with truncated downward jumps. The result is the desired forcing plus two explicit error terms, and the matrix makes both errors zero.

The corrected proof works for a concrete reason that can be checked line by line. Start with the interior candidate v_k . The rational construction guarantees

$$(q - \mathcal{L}^{(k),\text{rat}})v_k = G_k.$$

The true killed generator differs from the rational calculation only by the two truncation terms displayed in (20). Therefore substitution into the true equation gives

$$(q - \mathcal{L}^{(k),D})v_k = G_k + \lambda_{1,-}^{(k)} C_{r_{1,-}^{(k)}}[v_k] e^{-r_{1,-}^{(k)}x} + \lambda_{2,-}^{(k)} C_{r_{2,-}^{(k)}}[v_k] e^{-r_{2,-}^{(k)}x}.$$

The second and third matrix rows set the two C coefficients to zero, so the candidate satisfies the true killed PIDE on every $x > 0$. The first row gives $v_k(q, 0) = 0$, so it also satisfies absorption. Finally, uniqueness of the killed resolvent identifies this verified candidate with the probabilistic expectation. Unlike the old proof, no step changes the function being used below the barrier.

9 Translation back to the paper's notation

The original paper works in z and writes

$$c_{k,j}(h_0)e^{jz} = \binom{k}{j}(-b)^{k-j}e^{jh_0}e^{jz}.$$

At $x = z + h_0$, this becomes simply $d_{k,j}e^{jx}$. Evaluation at the initial state $z = 0$ corresponds to $x = h_0$, so

$$\widehat{U}_k(q, 0; h_0) = P_k(q, h_0) + \sum_{m=4}^6 B_m^{(k)}(q)e^{\gamma_m(q)h_0}. \quad (22)$$

This also shows that, after translation, the barrier coefficients $B_m^{(k)}$ do not depend on h_0 ; the initial buffer enters only through evaluation at $x = h_0$.

10 Replacement guidance for the manuscript

The revised paper should make the following changes.

1. Translate the barrier to zero with $x = z + h_0$ before deriving the resolvent.
2. Define the killed downward operator with upper limit x , as in (7).
3. Replace the claim “ $K_{d,-}[\widehat{U}](0^+) = 0$ and linearity gives conditions on the particular and homogeneous pieces” by Lemma 1 and the residual calculation (21).
4. Call $C_r[v]$ a truncation-correction coefficient. Do not identify it with the value of the killed jump operator at the boundary.
5. State the additional long-leg moment condition $k\eta_{1,+} < 1$.

6. Initially derive the time-Laplace transform for q to the right of its abscissa of convergence. Treat continuation to the complex Talbot contour as a separate numerical-analytic step.

Remark 1 (Coincident downward rates). If $r_{1,-}^{(k)} = r_{2,-}^{(k)}$, the two downward exponential kernels combine into a single phase. The two residual functions in (21) are then identical, so one should reduce the rational symbol and solve the lower-dimensional system. Describing this case only as a singular 3×3 matrix obscures the probabilistic reason for the degeneracy.

11 Conclusion

The original 3×3 coefficients can survive the correction, but the boundary proof cannot. The matrix rows involving $r/(r + \gamma)$ are cancellation conditions for barrier-truncation residuals, not values of the killed phase operator at the barrier. This distinction removes the extension inconsistency and remains valid for characteristic roots lying beyond a downward phase pole, where the full-line convolution interpretation is unavailable.